YITIAN WANG

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SUMMARY

Postdoctoral Fellow at The University of Texas at Austin with a Ph.D. in Electrical Engineering specializing in solid-state electrolytes for next-generation Li-ion batteries. Published 5 first-author papers on garnet, NaSICON, and perovskite solid electrolytes — three leading families in all-solid-state battery development. Experienced in crystal growth, cell assembly and testing, impedance spectroscopy, thermal transport characterization, and computational materials design. Led multi-institutional experiments at Oak Ridge National Laboratory coordinating across 7 groups at 5 institutions.

EDUCATION

University of California, Riverside, CA — *Mar. 2021 – Jun. 2026*
Ph.D. in Electrical and Computer Engineering
Awards: MRS 2023 Highly Commended Student Talk Award; Dissertation Completion Fellowship
Coursework: Solid-State Physics, Semiconductor Devices, Electrochemistry, Thin-Film Technology

Columbia University in the City of New York, NY — *Aug. 2019 – Feb. 2021*
Master of Science, Material Science and Engineering

Beijing Normal University, CN — *Sep. 2015 – Jun. 2019*
Bachelor of Science, Physics
Awards: 'Jingshi' Scholarship (2016 – 2018)

SKILLS

Cell Fab & Electrochemistry	Glovebox (Ar), Coin-Cell Assembly (CR2032), Separator Fabrication, EIS, Galvanostatic Cycling
Synthesis & Processing	Floating-Zone Crystal Growth, Solid-State Sintering, Powder Synthesis, Solution Processing, PVD, CVD
Characterization	XRD, SEM, TEM, AFM, Raman, FTIR, Inelastic Neutron Scattering, DSC, TGA, PPMS, Four-Probe Ionic/Electronic Transport
Software & Methods	Python, MATLAB, PyTorch, Quantum ESPRESSO, VASP, GSAS-II, Mantid, COMSOL, DOE, Statistical Analysis

RESEARCH AND ENGINEERING EXPERIENCE

Postdoctoral Fellow — *Jul. 2026 – Present*
The University of Texas at Austin — *Austin, TX*

- Conducting postdoctoral research in materials and energy-related engineering.

Graduate Research Assistant — Solid-State Electrolyte Development — *Jun. 2021 – Jun. 2025*
University of California, Riverside — *Riverside, CA*

- Synthesized and characterized three families of solid-state electrolytes (garnet LLZTO, NaSICON NZSP, perovskite LSHT), establishing structure–property–performance relationships across each system.
- Built a floating-zone crystal growth platform from scratch, producing centimeter-scale LLZTO single crystals — enabling phonon-resolved measurements previously infeasible on polycrystalline samples.
- Discovered *diffuson*-mediated thermal transport in Li-ion solid electrolytes, providing the first experimental verification of this mechanism and informing thermal management strategies for solid-state battery packs.
- Developed a two-channel thermal-conductivity model in MATLAB explaining anomalous temperature dependence where traditional Debye models failed — applied across all three electrolyte families.
- Performed comprehensive characterization (XRD, SEM, Raman, FTIR, DSC/TGA, four-probe) to evaluate phase purity, microstructure, and ionic/thermal transport properties.

Experiment Lead — Neutron Scattering on Solid Electrolytes — *Jan. 2022 – Jul. 2023*
Oak Ridge National Laboratory — *Oak Ridge, TN*

- Authored two successful beam time proposals and led inelastic neutron scattering experiments on LLZTO single crystals at ORNL (TAX/HFIR and ARCS/SNS).
- Resolved the full phonon dispersion of a garnet solid electrolyte for the first time, providing benchmarks for ab initio lattice-dynamics predictions of ionic conductivity.
- Built automated data-reduction pipelines in Mantid and Python, improving reproducibility and reducing analysis turnaround.
- Coordinated across 7 research groups at UCR, UT Austin, ORNL, ETH Zürich, and TU Wien.

Graduate Researcher — Battery Materials & Cell Testing *Sep. 2019 – Jan. 2021*
Columbia University *New York, NY*

- Engineered composite Li-ion battery separators (γ-C_3N_4/PVDF) via solution processing, optimizing film uniformity and mechanical properties for improved electrochemical kinetics.
- Assembled and tested CR2032 coin cells in an Ar-filled glovebox over 100+ galvanostatic cycles, characterizing capacity retention, rate capability, and interfacial stability via EIS.
- Applied first-principles DFT (Quantum ESPRESSO) to model defect stability in olivine cathode structures, guiding materials selection for electrode design.

PROFESSIONAL MEMBERSHIPS

- Sigma Xi, The Scientific Research Honor Society — Elected Full Member (2026)
- American Physical Society (APS) — Member (2025 – 2026)
- Materials Research Society (MRS) — Student Member (through 2027)

SELECTED PUBLICATIONS (5 FIRST-AUTHOR)

1. **Y. Wang** et al. "Low thermal conductivity of NaSICON-type solid electrolyte $Na_3Zr_2Si_2PO_{12}$." *Tungsten*, 2025. [Link]
2. **Y. Wang** et al. "Glass-like thermal transport in perovskite Li-ion conductor." *Chem. Commun.*, 2025 (Front Cover). [Link]
3. **Y. Wang** et al. "Origin of intrinsically low thermal conductivity in a garnet-type solid electrolyte." *PRX Energy*, 2025. [Link]
4. **Y. Wang** et al. "Thermal properties of Li-ion conducting garnet solid electrolyte LLZTO." *J. Mater. Chem. A*, 2024. [Link]
5. **Y. Wang**, X. Chen. "Single crystal growth of garnet-type fast Li-ion conductors." *Tungsten*, 2022. [Link]

MEDIA COVERAGE

- Why LLZTO is the Secret to Super-Fast EV Charging — UC Riverside, Feb. 2026
- Cool Battery Power — UC Riverside News, Oct. 2025
- ECE PhD Student Wins Materials Research Student Talk Award — UCR ECE, May 2023